

Chapter One

Introduction to Database Systems

Chapter 1 - Objectives

- **Some common uses of database systems.**
- **Characteristics of file-based systems.**
- **Problems with file-based approach.**
- **Meaning of the term database.**
- **Meaning of the term Database Management System (DBMS).**
- **Personnel involved in the DBMS environment.**
- **Advantages and disadvantages of DBMSs.**

Database systems

- **Today, Databases are essential to every business. They are used to maintain internal records, to present data to customers and clients on the World-Wide-Web, and to support many other commercial processes.**
- **Databases are likewise found at the core of many modern organizations.**

Examples of Database Applications

- **Purchases from the supermarket**
- **Purchases using your credit card**
- **Booking a holiday at the travel agents**
- **Using the local library**
- **Taking out insurance**
- **Renting a video**
- **E- Commerce**
- **Banking, Insurance and other Financial Systems**
- **Social media (in general the WWW)**

Data Handling Approaches in Organizations

- Data management passes through the different levels of development along with the development in technology and services. These levels could best be described by categorizing the levels into three levels of development. Even though there is an advantage and a problem to overcome at each new level, all methods of data handling are in use even today to some extent. The major three levels are;
 - **Manual Approach**
 - **Traditional File Based Approach**
 - **Database Approach**

Manual Approach

- Cards and paper are used for the purpose
- Files for as many event and objects as the organization has are used to store information.
- Each of the files containing various kinds of information is labelled and stored in one or more cabinets.
- The cabinets could be kept in safe places for security purpose based on the sensitivity of the information contained in it (Cabinet Lockers).
- Insertion and retrieval is done by searching first for the right cabinet then for the right the file then the information.
- One could have an indexing system to facilitate access to the data
- Limitations of the Manual approach
 - Prone to error
 - Difficult to update, retrieve, integrate
 - You have the data but it is difficult to compile the information
 - Limited to small size information
 - *Cross referencing is difficult*

File-Based Systems

- **Collection of application programs that perform services for the end users (e.g. reports).**
- **Each program defines and manages its own data.**

File-Based Processing

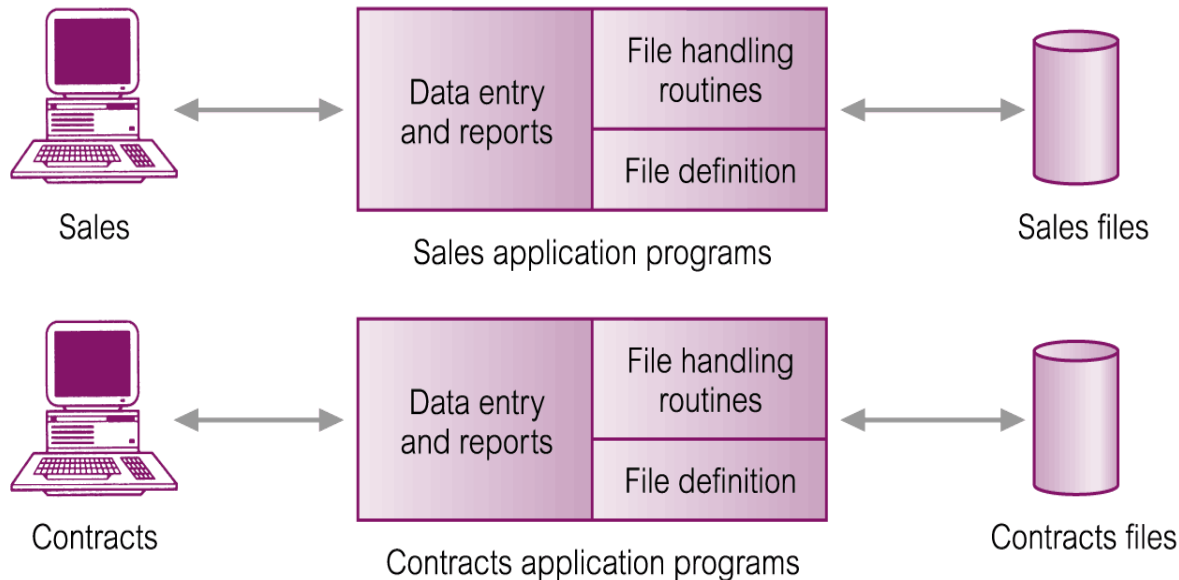


Figure 1.5

File-based processing.

Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)

PrivateOwner (ownerNo, fName, lName, address, telNo)

Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)

PropertyForRent (propertyNo, street, city, postcode, rent)

Client (clientNo, fName, lName, address, telNo)

Limitations of File-Based Approach

- **Separation and isolation of data**
 - **Each program maintains its own set of data.**
 - **Users of one program may be unaware of potentially useful data held by other programs.**
- **Duplication of data**
 - **Same data is held by different programs.**
 - **Wasted space and potentially different values and/or different formats for the same item.**

Limitations of File-Based Approach

- **Data dependence (Program Data Dependence)**
 - **File structure is defined in the program code.**
 - **Any change in the data structure need to changes the program as well**
- **Incompatible file formats**
 - **Programs are written in different languages, and so cannot easily access each other's files. (C, COBOL) – Data Structures are different for different Languages**
- **Fixed Queries/Proliferation of application programs**
 - **Programs are written to satisfy particular functions.**
 - **Any new requirement needs a new program.**

Database Approach

- **Problems of file approach arose because:**
 - **Definition of data was embedded in application programs, rather than being stored separately and independent of the applications.**
 - **No control over access and manipulation of data beyond that imposed by application programs.**
- **Solution(Result):**
 - **The database and Database Management System (DBMS).**
 - **This approach solves the problems of the File-based Approach**

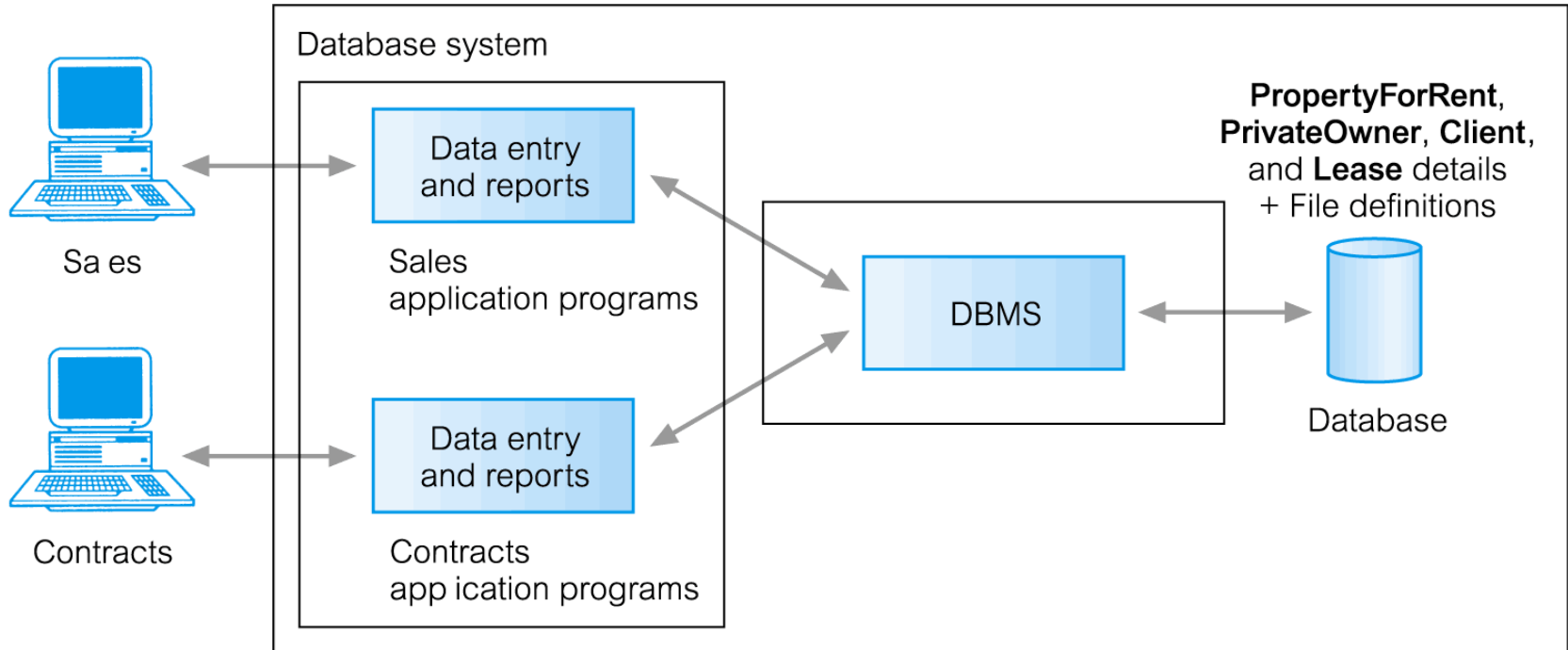
Database

- **Shared collection of logically related data (and a description of this data), designed to meet the information needs of an organization.**
- **System catalogue (metadata) provides description of data to enable program–data independence.**
- **Logically related data comprises entities, attributes, and relationships of an organization's information.**

Database Management System (DBMS)

- **A software system that enables users to define, create, maintain, and control access to the database.**
- **(Database) application program: a computer program that interacts with database by issuing an appropriate request (SQL statement) to the DBMS.**

Database Management System (DBMS)



PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)

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Database Approach Cont'd.....

- **Data definition language (DDL).**
 - Permits specification of data types, structures and any data constraints.
 - All specifications are stored in the database.
 - Enables the Creation, Alteration and Removal of a *Database Object*
- **Data manipulation language (DML).**
 - General enquiry facility (query language -retrieval) of the data.
 - In addition to querying, we can also have manipulation of Data (Adding New, Updating info, Deleting info)

Database Approach– Access Control

- **Controlled access to database may include: (User/Role- definition, Privilege Assignment/Revocation, Access Enforcement)**
 - a security system
 - an integrity system
 - a concurrency control system
 - a recovery control system
 - a user-accessible catalogue.

Database Views

- **The Database Approach introduces a little complexity on part of the end user.**
- **Although the Database is a shared collection, users are interested in their specific data needs**
- **A view allows each user to have his or her own view of the database.**
- **A view is essentially some subset of the database.**
- **Data Irrelevant to a user is not at all visible**

Views - Benefits

- **Reduce complexity**
- **Provide a level of security**
- **Provide a mechanism to customize the appearance of the database**
- **Present a consistent, unchanging picture of the structure of the database, even if the underlying database structure is changed**

Roles in the Database Environment

- **Data Administrator (DA)**
 - Responsible on management of data resources. This involves in database planning, development, maintenance of standards policies and procedures at the conceptual and logical design phases.
- **Database Administrator (DBA)**
 - This is more technically oriented role. DBA is responsible for the physical realization of the database. It is involved in physical design, implementation, security and integrity control of the database.
 - Also deals with Optimizing the performance of the system

Database Designers

- **Database Designers (Logical and Physical)**
 - Identifies the data to be stored and choose the appropriate structures to represent and store the data.
 - Should understand the user requirement and should choose how the user views the database.
 - Involve on the design phase before the implementation of the database system.
- **We have two distinctions of database designers, one involving in the logical & conceptual design and another involving in physical design.**

Database Designers- Cont'd...

- **Logical and Conceptual DBD**

- Identifies data (entity, attributes and relationship) relevant to the organization
- Identifies constraints on each data
- Understand data and business rules in the organization
- Sees the database independent of any data model at conceptual level and consider one specific data model at logical design phase.

- **Physical DBD**

- Take logical design specification as input and decide how it should be physically realized.
- Map the logical data model on the specified DBMS with respect to tables and integrity constraints. (DBMS dependent designing)
- Select specific storage structure and access path to the database
- Design security measures required on the database

Application Programmers

- System analyst determines the user requirement and how the user wants to view the database.
- The application programmer implements these specifications as programs; code, test, debug, document and maintain the application program.
- **The application programmer determines the interface on how to retrieve, insert, update and delete data in the database.**

End Users (naive and sophisticated)

- **Naïve Users:**

- Sizable proportion of users
- Unaware of the DBMS
- Only access the database based on their access level and demand
- Use standard and pre-specified types of queries.

- **Sophisticated Users**

- Users familiar with the structure of the Database and facilities of the DBMS.
- Have complex requirements
- Have higher level queries
- Are most of the time engineers, scientists, business analysts, etc

- **Casual Users**

- Users who access the database occasionally.
- Need different information from the database each time.
- Use sophisticated database queries to satisfy their needs.
- Are most of the time middle to high level managers

Advantages of DBMSs

- **Control of data redundancy**
- **Data consistency**
- **More information from the same amount of data**
- **Sharing of data**
- **Improved data integrity**
- **Improved security**
- **Enforcement of standards**
- **Economy of scale**

Advantages of DBMSs

- **Balance conflicting requirements**
- **Improved data accessibility and responsiveness**
- **Increased productivity**
- **Improved maintenance through data independence**
- **Increased concurrency**
- **Improved backup and recovery services**

Disadvantages of DBMSs

- **Complexity**
- **Size**
- **Cost of DBMS**
- **Additional hardware costs**
- **Cost of conversion**
- **Performance**
- **Higher impact of a failure**